

Jasper Chapman

Bachelor of Science in Mechanical Engineering, Additional Major in Robotics | Class of 2029

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Education

Carnegie Mellon University — Pittsburgh, PA

Expected May 2029

Bachelor of Science in Mechanical Engineering; Additional Major in Robotics

Relevant Coursework: Human-Robot Interaction, Fundamentals of Mechanical Engineering, Electronics/Sensors & Actuators; Thermodynamics, Mechanics I & CNC Machining (in progress)

Technical Skills

CAD & Simulation: SOLIDWORKS, Fusion 360, Onshape, Blender

Programming: Python, C, Java, Arduino (embedded firmware)

Manufacturing: Carbon-fiber / prepreg composite layup, materials testing, manual machining, 3D printing, laser cutting

Software & Tools: Figma, Adobe Premiere Pro, Microsoft Office, Google Workspace, Slack

Experience & Research

Carnegie Mellon Robotics Institute — Zoom Lab (Dr. Zeynep Temel)

May 2026 – Present

Research Assistant

Pittsburgh, PA

- Designing a palm-scale morphing quadrotor to fold its arms and switch between flight and tank-tracked ground locomotion via passive spring mechanics and prop thrust, with no dedicated actuator.
- Collaborating with lab researchers to model mechanisms in Onshape and integrate flight firmware in Python.

MIT AgeLab

June 2024 – August 2024

Research Intern

Cambridge, MA

- Built AI-assisted reminiscence-therapy tools for early-onset Alzheimer's patients (Python, GPT-4, sentiment analysis); presented findings to a panel of MIT researchers and faculty.

Projects

Solar-Powered Off-Grid Aquaculture System

September 2023 – February 2024

Independent Engineering Project — 1st Place, PA Junior Academy of Science

- Engineered a solar-powered, off-grid aquaculture system to give a rural Sierra Leone mining community self-sustaining food production; earned 1st place out of 2,000+ project entries.

Koi For Good — Nonprofit Initiative

March 2019 – May 2024

Founder

koiforgood.com

- Founded and led a nonprofit over 5 years, selling 300+ koi to raise \$8,600 and furnish classrooms for 400 schoolchildren across rural Sierra Leone.

Leadership & Activities

Carnegie Mellon Racing (Formula SAE)

September 2025 – Present

Monocoque Team Member — SES Panel Testing System Lead

Pittsburgh, PA

- Lead SES panel strength testing with a 150-member team and hand-lay carbon-fiber prepreg to certify material compliance for CMU's Formula SAE competition vehicle.

Teaching Assistant — Design Thinking with AI (70-389), Carnegie Mellon

January 2026 – Present

Selected by faculty to support instruction for a 50-student applied-design and AI course.

Pittsburgh, PA

Awards & Honors

- Parkin Fellowship for Global Service** (2024) — Awarded \$6,300 to research conflict-mineral sites across Sierra Leone.
- Cum Laude Society** — Inducted for ranking in top 20% of a 130-student class across academics and citizenship.
- AIME Qualifier** (2024) — Sole student in graduating class to qualify (top ~5% of AMC participants nationally).

Extracurriculars & Interests

Athletics & Music: Varsity Swimming (WPIAL & PA State Championships), Varsity Track & Field (discus, 7th all-time), CMU Club Water Polo; Cello (School Orchestra)

Service & Societies: Youth Mentor, Root Studios Sierra Leone; 1-on-1 STEM tutoring; ASME, Explorers Club, Co-President of the World Cinema Club